INDIAN STATISTICAL INSTITUTE
Mathematics-3 (BSDS)
QUIZ -1 Examination

DATE: 12/09/2025 **Total Marks:** 15 **Time:** 4:15 – 5:15 pm

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MARKS WILL BE AWARDED ONLY FOR THOSE ANSWERS WITH PROPER JUSTIFICATION

Question 1: Let V be a complex inner product space and let $T : V \to V$ be a linear transformation. Show that $T \equiv 0$ if and only if $\langle Tv, v \rangle = 0$ for each $v \in V$. Is this result true for real inner product space. [5]

Question 2: Orthogonally diagonalize

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}.$$

[5]

Question 3: Compute the SVD of

$$A = \begin{bmatrix} 1 & -1 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}.$$

[5]

— ALL THE BEST —